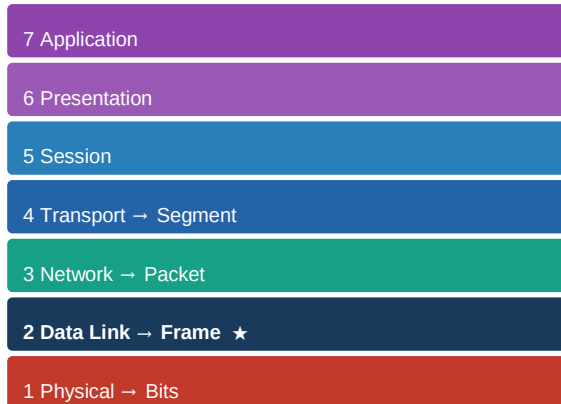


Ethernet LAN Switching

Topics Covered: OSI Layers 1 & 2 • LAN Definition • Ethernet Frame • MAC Addresses • Hexadecimal • Switch Operation • Multi-Switch Scenarios • Quiz Review

1. OSI Model Review

OSI MODEL



Layer 1 – Physical Layer

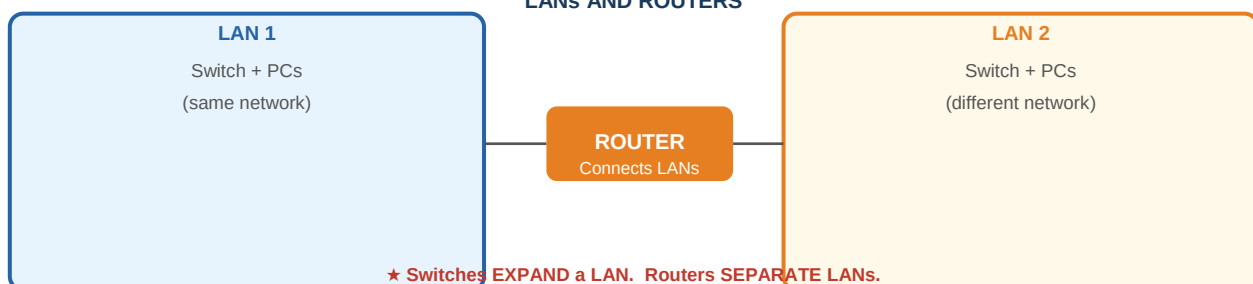
- Defines physical characteristics: voltage, max distances (UTP = **100m**), connectors, specs.
- Converts digital bits → electrical (wired) or radio (wireless) signals.
- Covered: copper UTP cables, fiber-optic, RJ45 connectors.

Layer 2 – Data Link Layer

- Node-to-node connectivity: PC↔Switch, Switch↔Router, Router↔Router.
- Defines data format for transmission; detects (possibly corrects) Physical Layer errors.
- Uses **Layer 2 MAC addressing** — separate from Layer 3 IP addressing.
- **Switches operate at Layer 2.** Ethernet covers both L1 and L2.

2. What Is a LAN?

LANs AND ROUTERS

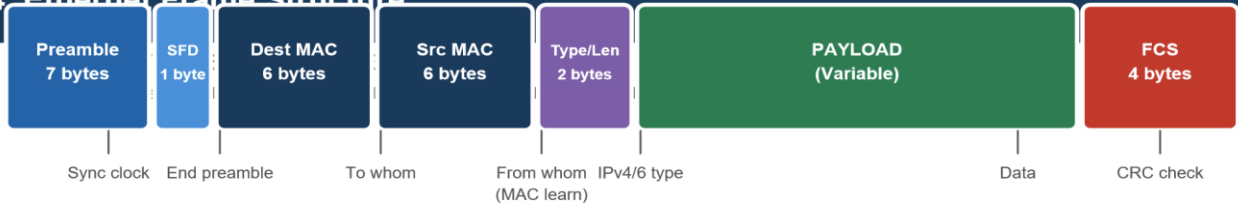


- **LAN (Local Area Network):** network within a small area (office, home).
- Routers connect *separate* LANs. Switches *expand* an existing LAN.
- Two switches connected = still ONE LAN. Switch connected to different router interfaces = TWO LANs.

3. PDUs & Encapsulation

Layer	PDU Name	What Gets Added
Upper Layers (5-7)	Data	—
Layer 4 – Transport	Segment	L4 Header
Layer 3 – Network	Packet	L3 Header
Layer 2 – Data Link	Frame ★	L2 Header + Trailer
Layer 1 – Physical	Bits	Encoded as signals

4. Ethernet Frame Structure



Total header+trailer = 26 bytes | Preamble+SFD not counted in official Ethernet frame size

Field Details

Field	Size	Bit Pattern / Value	Purpose
Preamble	7 bytes 56 bits	10101010 × 7	Synchronize receiver clock
SFD	1 byte 8 bits	10101011	End of preamble / start of frame

Destination MAC	6 bytes 48 bits	Target MAC address	Who should receive this frame
Source MAC	6 bytes 48 bits	Sender MAC address	Used by switch to LEARN MAC → port
Type / Length	2 bytes 16 bits	≤1500 = length ≥1536 = type	0x0800 = IPv4 0x86DD = IPv6
FCS (trailer)	4 bytes 32 bits	CRC result	Detect transmission errors
TOTAL H+T	26 bytes	—	Preamble+SFD not in official frame size

0x0800 = 2048 decimal → IPv4 | 0x86DD = 34525 decimal → IPv6 | 0x prefix = hexadecimal notation

FCS uses **CRC (Cyclic Redundancy Check)**: Cyclic=cyclic codes, Redundancy=adds no new info, Check=verifies data.

5. MAC Addresses

MAC ADDRESS ANATOMY (example: E8BA.7011.2874)

E8BA.70

OUI — First 3 bytes (24 bits)

Assigned to MANUFACTURER

Total: 6 bytes = 48 bits | Written as 12 hex chars | Also called BIA (Burned-In Address) | Globally unique

11.2874

Device ID — Last 3 bytes (24 bits)

Unique to THIS DEVICE

- **MAC = Media Access Control** — 6-byte (48-bit) physical address.
- Assigned at manufacture; also called **BIA (Burned-In Address)**.
- Different from IP (logical, software-assigned). MAC is hardware-level.
- **Globally unique** — no two devices worldwide share the same MAC.
- Written as 12 hex characters, e.g. **AAAA.AA00.0001** (periods every 4 chars) or AA:AA:AA:00:00:01.

6. Hexadecimal

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

HEXADECIMAL QUICK REFERENCE

Hex: 0 1 2 3 4 5 6 7 8 9 A B C D E F

Dec: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

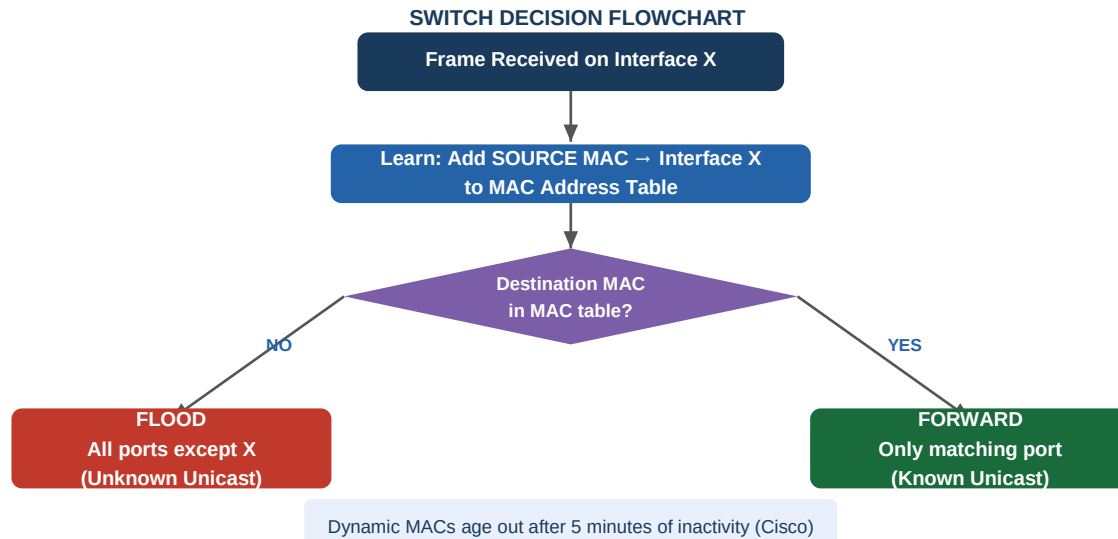
Next column = ×16 | 0x prefix = hexadecimal | MAC = 12 hex digits = 48 bits

Letters represent numbers !

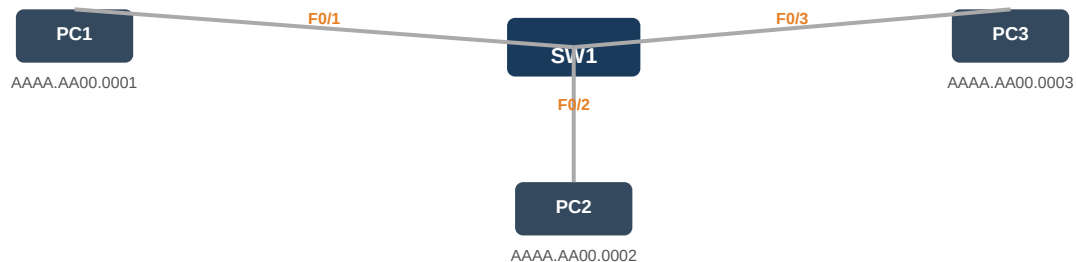
- Decimal = 10 digits (0–9). Hexadecimal = **16 digits** (0–9, A–F).
- Each hex column represents a power of **16** (vs. 10 in decimal).
- Hex 10 = decimal 16. Hex 1A = decimal 26. Hex FF = decimal 255.
- Prefix **0x** indicates a hexadecimal number (e.g. 0x0800).

- IPv6 uses hex extensively — covered in a later lesson.

7. Switch MAC Address Learning & Forwarding



Network: SW1 with PC1, PC2, PC3



Core Rules

Rule	Detail
Learn from SOURCE	Read source MAC of every received frame → add to MAC table with incoming interface
Interface meaning	MAC table interface = 'use this port to reach this MAC' (not necessarily direct connection)
Unknown Unicast → FLOOD	Dest MAC not in table → send out ALL ports except receiving port
Known Unicast → FORWARD	Dest MAC IS in table → send only out the matching port
5-Minute Aging	Cisco switches remove dynamic MACs after 5 min of inactivity
No loop-back	Never flood back out the interface the frame arrived on

Step	Action	Event
1	PC1 sends unicast frame to PC2.	SW1 receives on F0/1
2	SW1 reads source MAC (PC1) → learns: PC1 = F0/1	MAC table updated

Step-by-Step: PC1 → PC2 (Single Switch)

3	PC2's MAC not in table → Unknown Unicast	FLOOD out F0/2, F0/3
4	PC3 drops frame (wrong dest MAC). PC2 processes it.	PC2 receives
5	PC2 replies to PC1. SW1 receives on F0/2 → learns PC2 = F0/2	MAC table updated
6	PC1 MAC is known (F0/1) → Known Unicast → FORWARD to F0/1	No flood

8. Two-Switch Scenario

Two-Switch Network: PC1→ PC3 Path



Step-by-Step: PC1 → PC3 (Two Switches)

Step	Action	Learned/Result
1	PC1 → frame to PC3. Arrives at SW1 on F0/1.	SW1 learns PC1 → F0/1
2	PC3 unknown to SW1 → Flood out F0/2 & F0/3.	SW1 floods
3	SW2 receives frame on F0/3. Learns PC1 → F0/3 (path via SW1).	SW2 learns PC1
4	PC3 unknown to SW2 → Flood out F0/1 & F0/2.	SW2 floods
5	PC4 drops (MAC mismatch). PC3 receives and processes.	PC3 receives
6	PC3 replies. SW2 receives on F0/1 → learns PC3 → F0/1.	SW2 learns PC3
7	SW2 checks dest (PC1): found → F0/3. Forward (no flood).	SW2 forwards
8	SW1 receives on F0/3 → learns PC3 → F0/3.	SW1 learns PC3
9	SW1 checks dest (PC1): found → F0/1. Forward to PC1.	Delivered!

Final MAC Address Tables

Switch	MAC Address	Interface	Learned When
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SW1	PC1 — AA..0001	F0/1	PC1's first frame (Step 1)
SW1	PC3 — AA..0003	F0/3	PC3's reply passes through SW2→SW1 (Step 8)
SW2	PC1 — AA..0001	F0/3	PC1's frame arriving from SW1 (Step 3)
SW2	PC3 — AA..0003	F0/1	PC3's reply, source MAC on F0/1 (Step 6)

9. Quiz Review (5 Questions)

Q1: Which field provides receiver clock synchronization?

✓ **Correct: A – Preamble**

7 bytes of 10101010 allow the device to synchronize its receive clock. SFD signals end of preamble. Type = L3 protocol. FCS = error detection.

Q2: How long is a MAC address?

✓ **Correct: D – 48 bits**

6 bytes × 8 bits = 48 bits. IP address = 32 bits. 48 bytes would be 384 bits — a common trap.

Q3: What is the OUI of E8BA.7011.2874?

✓ **Correct: B – E8BA.70**

OUI = first 3 bytes (24 bits / 6 hex chars) = E8BA.70. Assigned to the manufacturer. Remaining 3 bytes = device-specific.

Q4: Which field does a switch use to populate its MAC table?

✓ **Correct: C – Source MAC Address**

The switch reads the SOURCE MAC and maps it to the incoming interface. Destination MAC is used for forwarding decisions, not learning.

Q5: What kind of frame is flooded out all ports except the receiving one?

✓ **Correct: A – Unknown Unicast**

Unknown unicast = single destination but MAC not in table → flood. Known unicast = MAC found → forward. 'Allcast' is not a valid frame type.

10. Key Terms Glossary

Term	Definition
LAN	Local Area Network — small geographic network
MAC Address	6-byte/48-bit burned-in physical address (BIA)
OUI	First 3 bytes of MAC — manufacturer identifier
BIA	Burned-In Address — hardware MAC address
PDU	Protocol Data Unit — data at each OSI layer
Frame	Layer 2 PDU — header + payload + trailer
FCS	Frame Check Sequence — 4-byte CRC error detection trailer
CRC	Cyclic Redundancy Check — error detection algorithm
Preamble	7-byte sync pattern (10101010 ×7)
SFD	Start Frame Delimiter — 1 byte, marks frame start (10101011)

Unicast Frame	Frame destined for a single device
Unknown Unicast	Dest MAC not in table → switch floods
Known Unicast	Dest MAC in table → switch forwards to specific port
Flood	Send frame out all interfaces except the one it arrived on
Dynamic MAC	Auto-learned by switch; expires after 5 min (Cisco)
Hexadecimal	Base-16 numbering (0–9, A–F); used for MAC & IPv6

★ Key Mantra: LEARN from Source MAC | FLOOD Unknown Unicast | FORWARD Known Unicast | Age out = 5 min